GUJARAT TECHNOLOGICAL UNIVERSITY

Program Name: Diploma Engineering

Level: Diploma

Branch: Plastics Engineering

Subject Code : DI05023021

Subject Name : Recycling of Plastics

w. e. f. Academic Year:	2026-27
Semester:	5th
Category of the Course:	PCC

Prerequisite:	Processing subjects & Material Subjects
Rationale:	Plastics are being used in practically all areas of consumer products, including construction, transportation, packaging, automobile and agriculture. In recent world more concern is on the possible damaging impact of plastics on the environment. A Plastic Diploma engineer has to identify all possible sources of plastic waste generation and do recycling of it with or without energy recovery within government norms. This competency requires the knowledge of various plastic recycling techniques. Hence the course has been designed to develop this competency and its associated cognitive, practical and affective domain learning outcomes.

Course Outcome:

After Completion of the Course, Student will able to:

No	Course Outcomes	RBT Level
01	Select waste disposal method.	R,U,A
02	Select proper separation method.	R,U,A
03	Identify primary and secondary recycling methods.	U,A
04	Analyze various tertiary and quaternary recycling methods.	U,A
05	Understand recycling methods of various plastics.	R,U

**Revised Bloom's Taxonomy (RBT)*

Teaching and Examination Scheme:

Teaching Scheme (in Hours)			Total Credits L+T+ (PR/2)	Assessment Pattern and Marks				Total Marks
				Theory		Tutorial / Practical		
L	T	PR	C	ESE(E)	PA(M)	PA(I)	ESE (V)	
3	0	2	4	70	30	20	30	150

Course Content:

Unit No.	Content	No. of Hours	% of Weightage
1.	**Sources of Plastic Waste And Its Management** 1.1 Introduction to Plastics Waste 1.2 Definitions of related terms – Waste Plastic, Industrial Plastic Waste, Postconsumer Plastic Waste, Nuisance Plastic, Scrap Plastic, Primary Recycling, Secondary Recycling, Tertiary Recycling And Quaternary Recycling 1.3 Economic and environmental impact of Plastic Waste 1.4 Sources of waste – postconsumer, municipal and industrial waste 1.5 Management of plastic waste - 4 R (reduction, reuse, recycling and recovery) 1.6 Plastic cycle - flow of plastic products and plastic waste 1.7 Waste disposal a. Landfill of municipal solid waste– open dumping and sanitary landfill b. Plastics in landfill c. Future trends	9	20% (14 Marks)
2.	**Separation Methods** 2.1 Size reduction by mechanical methods 2.2 Separation of plastics using physical properties 2.3 Separation using recycling codes 2.4 Separation processes specific to plastics a. Separation of paper/plastic mixtures Application of Heat, Wet Separation Process, Electro Dynamic Separation b. Separation of plastic from plastic-coated fabric c. Separation of mixtures of plastics-Float Sink Method, Using Selective Wetting Characteristics, Solvent Separation	9	20% (14 Marks)
3.	**Primary & Secondary Recycling Methods** **Primary Recycling** 3.1 Introduction 3.2 Primary recycling methods a. Granulators b. Cryogenic grinding c. Plunger and screw type stuffer d. In-line recycling **Secondary Recycling**	9	20% (14 Marks)

	3.3 Introduction to secondary recycling 3.4 Various technical approaches for secondary recycling 3.5 Secondary recycling by mechanical reworking of plastic waste 3.6 Recycling by chemical modification of plastic waste 3.7 Secondary recycling by co-extrusion and co-injection molding 3.8 Use of plastic as a filler		
4.	**Tertiary And Quaternary Recycling** **Tertiary Recycling** 4.1 Pyrolysis a. Introduction to pyrolysis and its advantages b. Introduction to pyrolysis reactors of plastics waste – Union Carbide System, Reactor by Japan Steel Works 4.2 Chemical decomposition of plastic waste a. Hydrolysis b. Glycolysis **Quaternary Recycling** 4.3 Introduction to quaternary recycling 4.4 Constructional features of incinerators 4.5 Incineration of plastic waste and its problems	9	20% (14 Marks)
5.	**Recycling Methods Of Specific Plastics** 5.1 Describe recycling methods of following plastics : a. PVC b. PET c. PMMA d. HDPE e. LDPE f. PS	9	20% (14 Marks)
	Total	**45**	**100** **(70 Marks)**

Suggested Specification Table with Marks (Theory):

Distribution of Theory Marks (in %)					
R Level	**U Level**	**A Level**	**N Level**	**E Level**	**C Level**
19	**47**	**04**	**------**	**------**	**------**

Where R: Remember; U: Understanding; A: Application, N: Analyze and E: Evaluate C: Create (as per Revised Bloom's Taxonomy)

References/Suggested Learning Resources:

(a) Books:

Sr. No.	Title of Book	Author	Publication with place, year and ISBN
1.	Plastic Waste	Jacob Leidner	Publication: New York : M. Dekker Publication Year: 1981 ISBN : 9780824713812
2.	Feedstock Recycling of Plastic Waste	Jose Aguado & David Serrano	Publication: Royal Society of Chemistry Publication Year: 1999 ISBN : 9780854045310
3.	Mixed Plastic Recycling Technology	Bruce Hegberg, Gary Brenniman W.H. Hallenback	Publication: New York : M. Dekker Publication Year: 1992 ISBN : 9780815518389
4.	Plastics Fabrication and Recycling	Manas Chanda, Salil K. Roy, Donald E. Hudgin	Publication: CRC Press LLC Publication Year: 2008 ISBN : 9781420080629
5.	Recycling of Plastic Materials	La Mantia	Publication: ChemTec Publishing Publication Year: 1993 ISBN : 9781895198034
6.	Modern Plastics Handbook	Harper	Publication: McGraw-Hill Professional Publication Year: 2000 ISBN : 9780070267145
7.	Plastics waste - feedstock recycling chemical recycling and incineration	Tukker	Publication: New York : M. Dekker Publication Year: 1981 ISBN : 9781859573310

8.	Recycling of PVC and Mixed Plastics Wastes	La Mantia, F.P. (ed)	Publication: ChemTec Publishing Publication Year: 1981 ISBN : 9781895198119

(b) Open source software and website:

1. http://www.g.eng.cam.ac.uk/impee/topics/RecyclePlastics/files/Recycling%20Plastic%20v3%20PDF.pdf
2. https://enterclimate.com/blog/methods-of-plastic-waste-management/
3. http://plasticisrubbish.com/2013/03/20/recycling-plastic-2
4. https://www.icpe.in/Plastics%20in%20Food%20Packaging/pdf/14-Final.pmd.pdf
5. https://web.anl.gov/PCS/acsfuel/preprint%20archive/Files/Volumes/Vol42-4.pdf
6. https://en.wikipedia.org/wiki/Plastic_recycling
7. https://www.ecomena.org/recycling-pvc/
8. https://www.inciner8.com/blog/waste-management/the-incinerators-guide-to-plastics
9. https://www.youtube.com/watch?v=tbpjX7GLzMc
10. https://www.ncbi.nlm.nih.gov/books/NBK233614/
11. https://pubs.acs.org/doi/10.1021/acssuschemeng.1c05013
12. https://ebooks.inflibnet.ac.in/esp11/chapter/recycling/

Suggested Course Practical List:

Sr. No.	Practical Outcomes (PrOs)	Unit No.	Approx. Hrs. required
1	To study various sources of plastic waste.	I	02
2	To study waste disposal methods.	I	02
3	To perform plastic waste size reduction by mechanical method.	II	02
4	To carry out separation of plastics using float-sink method.	II	02
5	To study various separation methods of paper/plastic mixtures.	II	02
6	To perform primary recycling of plastic using granulators.	III	02
7	To study recycling by chemical modification of plastic waste.	III	02
8	To study secondary recycling by co-extrusion and co- injection moulding.	III	02
9	To study various pyrolysis reactors.	IV	02
10	To study quaternary recycling methods.	IV	02
11	To carry out depolymerisation of PMMA for recovery of monomer.	V	02
12	To study recycling methods of PET	V	02
13	To study recycling methods of PVC.	V	02
14	To study recycling methods of HDPE.	V	02

List of Laboratory/Learning Resources Required:

Sr. No.	Equipment Name with Broad Specifications
1	Granulators
2	Crushers, shredder, hammer mill, disk mill
3	Electro dynamic separators, wet pulping separator
4	Sink float separator
5	Pyrolysis reactors
6	Air circulating incinerators
7	Laboratory type small extruder
8	Co-injection Machine
9	Co-extrusion Machine

Suggested Project List:

a) Prepare a chart for different recycle plastics
b) Create a paper recycling station to make new school supplies.
c) Construct benches from eco-bricks made of recycled plastic for outdoor seating.
d) Host a community art project using recycled materials.
e) Build a greenhouse using collected plastic bottles for the school garden.
f) Arrange clean-up events to collect plastic and paper waste, promoting environmental awareness and community involvement
g) Prepare report on Technical Process Flow Chart for Hard Plastic Waste Recycling
h) Prepare presentation on workshops for crafting with recycled plastic and paper.
i) Organize a fashion show featuring outfits made from recycled materials.
j) Collect products made up of co-extrusion/co-injection moulding process.
k) Make model for Recycle, reduce, and recover of waste plastics.
l) Prepare report on various pyrolysis recycling of plastic processes.

Suggested Activities for Students:

1. Students will read various papers on internet regarding various recycling processes.
2. Students will collect different plastic products with recycling codes.
3. Students will prepare identification chart with recycling codes.
4. Students will prepare flow chart of plastic cycle.
5. Students will visit nearby industries and collect information about recycling process.

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